

ARITHMETIC AND CALIBRATION

CI head 417f261 / merge 63e9887 | Verilator 5.020 | Group 1/3

22 RTL benches passed; independent arithmetic audit passed

RTL CI job succeeded; workflow conclusion is separate

BENCH / OBSERVED LOG COUNTS

REFERENCE CHECK / COVERAGE BOUNDARY

p2_oracle_tb

1,048,576 input codes; 0 errors

Observed: P2_ORACLE_COMPLETE codes=1048576 errors=0

Identity: dout == input code;
zero flags and 11-cycle completion.
One-slice fixture; not all configurations.

divider_borrow_tb

67,592 operand/config combinations

6 width/unroll fixtures

Observed: DIVIDER_BORROW_COMPLETE exhaustive_checks=67592

Signed floor reference + defining
inequality; zero divisor, busy and hold.
Six small-width parameter fixtures.

cal_weight_reduce_ppa_tb

1,430,848 miter input cases / 7 fixtures

43,758 allocations in 18-slice fixture

Observed: CAL_WEIGHT_REDUCE_PPA_COMPLETE

Independent physical-cell scalar sum;
old + uncached + cached T/G/R and IDs.
Transaction-edge sampling; no mapped/STA claim.

weight_row_cache_tb

16,150 loader transactions / 5 geometries

all words + exact cached row totals checked

Observed: WEIGHT_ROW_CACHE_COMPLETE geometries=5 steps=16150

Independent retained-word row sums;
reset/clear/write, completion/readback.
Five loader geometries; no measured PPA claim.

calibration_physical_tb

2,048 transactions; 18 physical slices

Observed: CALIBRATION_PHYSICAL_COMPLETE

Independent 256-bit integer formula;
code, flags, ID and input capture.
18 slices, reduced 8-cell test geometry.

calibration_recovery_tb

2,048 target/residue transactions

max |error|: calibrated 4; nominal 504 codes

Observed: CALIBRATION_RECOVERY_COMPLETE

Known target -> synthetic residue ->
calibrated and nominal-weight outputs.
Synthetic mismatch + ideal ADC2 quantization.

calibration_fit_tb

128 holdout samples; 1,278 weights

code + flags + ID + 11-cycle latency

Observed: CALIBRATION_FIT_COMPLETE

Frozen external holdout vectors;
code/flags/ID/latency + coefficient readback.
Fitted coefficients loaded, not learned in RTL.

p2_tb

23,049 TB checks; 0 errors

Observed: P2 RESULT: PASS

Golden vectors + floor inequality;
rounding, masks, carry, clipping and link.
T5 full-code sweep is a separate bench.

Counts retain their own units. No combined 'total coverage' score. All 22 log and TB hashes verified.

Source: docs/evidence/20260930/ci_rtl_417f/ | RTL only; not STA or analog performance.